TABLE OF CONTENTS

CURRICULUM VITAE

NAME: HUONG THI THANH HA

POSITION TITLE: Lecturer (similar to Assistant Professor role in the US system)

EDUCATION/TRAINING

Institution and Location	Degree	Completion Date	Field of Study
EBRAINS & IBRO		7/2021	Neuroscience (First Virtual Master Class on Brain Atlasing and Simulation Services)
France – Vietnam training for neurologists	Certificate	12/2019	Neuroscience
University of Medicine and Pharmacy, Ho Chi Minh City, Vietnam	Certificate	10/2019	Diagnostic test standard and procedure at hospitals.
Stanford University School of Medicine, Stanford, CA, US	Ph.D.	8/2018	Neuroscience
The Marine Biological Laboratory, MA, US	Certificate	8/2014	Neurobiology
University of Science, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	B.S.	09/2011	Medical biotechnology

A. Personal Statement and Research Interest

Due to a family history of multiple members with mental disorders, I became intrigued by how the brain works, what molecules drive its functions, and how they changed when it gets dysregulated. During college, I studied molecular markers for devising diagnostic tests of cervical cancer. Following graduation, I further pursued this interest by working at the Oxford University Clinical Research Unit and investigating peripheral blood markers of HIV-associated neurological disorders (HAND). At Stanford University, my PhD research was focused on understanding the molecular system related to autism and cellular and network deficits underlying epilepsy. I also created an efficient gene delivery technology and a universal brain tissue culture platform for research and drug discovery applications.

Upon coming back to Vietnam and joining the International University as a lecturer, I established the Brain Health Lab and pursue several research projects to address mental health issues and cognitive-aging disorders in Vietnam. In terms of brain health, my primary focus is on enhancing diagnoses and interventions for dementia at the early stage. My research encompasses a range of aspects, including molecular diagnosis and AI applications. For molecular detection, I am leading early diagnosing AD, focusing on detecting p-Tau217 and cell-free RNA detection. For studies related to AI, I am developing an easy-to-use AI system to detect early signs of Alzheimer's Disease (AD) in clinical settings or the biological characteristics of the heterogeneity in the Mild Cognitive Impairment (MCI) population. For the intervention aspect, I am building a cognitive game for MCI patients to hinder their conversion to dementia and evaluate the efficacy of EEG signals in predicting post-intervention effectiveness. Besides dementia research, my studies also apply AI to build a Brain-Computer Interface system for stroke rehabilitation and investigate AI potential in detecting brain tumors and migraines. Along with those studies, I am leading projects to build the first Vietnamese databases of neuroscience-related data, such as brain MRI and EEG signals to enhance AI models and improve diagnosis. In the realm of mental health, my research delves into the relationship between stress and emotions based on changes in EEG signals, as well as the efficacy of techniques such as Alpha music and Mindfulness-Based Stress Reduction in reducing stress. Moreover, I am also looking into developing a low-cost, non-invasive point-of-care device to quantify cortisol for stress diagnosis. My goal is to create innovative, highly accurate, and low-cost products that can have a tangible impact on the brain health and mental well-being of those in my community and beyond.

B. Publications, Laboratory products, Presentations and Posters

<u>Publication</u>

2025

Feasibility of home blood pressure telemonitoring in older adults with Mild Cognitive Impairment and uncontrolled hypertension in Vietnam (Authors: Vy Le Tran, Tran Tran To Nguyen, The Ha Ngoc Than, Dinh Chau Bao Hoang, Tuan Chau Nguyen, Viet Ngoc Tran, Toi Van Vo, Nam Phuong Nguyen, **Huong Thi Thanh Ha**) (Accepted, Biomedical Research and Therapy)

Assessing the feasibility of cognitive intervention via BrainTrain cognitive gaming application for Mild Cognitive Impairment patients (Authors: Thu Ngoc Minh Phan, Binh Nguyen Le Nguyen, Bao Quoc Nguyen, Nhung Hoang Do, Nhu Thi Huynh Tran, Nhan Thanh Nguyen, Thu Hoang Anh Pham, Nhan Quoc Trung Nguyen, Tuong Huu Nguyen, Kien Trung Duong, Nghia Tien Trong Hoang, Thu Thi Hoai Tran, Dang Minh Ly, and Dieu Xuan Nguyen Tan Le-Duy, and **Huong Ha Thi Thanh**) (Accepted, Int. J. Bioinformatics Research and Applications)

Ma Thi, C., Kien Nguyen, M., Vu Thanh, L., Nguyen Dinh, H., Huynh Thi, N. Y., **Ha, H. T.**, ... & Le Hoang, T. L. UET175: EEG dataset of motor imagery tasks in Vietnamese stroke patients. Frontiers in Neuroscience, 19, 1580931.

Ngo, L., Le, A., Ho, K., Le, T., Nguyen, N., Nguyen, S., ... & **Ha, H.** (2025). Electroencephalogram (EEG) based automated detection of mental disorders using artificial intelligence processing pipelines. Neuroscience Research Notes, 8(2), 404-1.

Tran, V. Q. T., Tran, N. T. L., Nguyen, L. Q., An, H. M. A., Nguyen, T. H., Trang, H. L., ... & **Ha, T. T. H.** (2025). Effect of Alpha Music on the Stress Level of Vietnamese Students. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 363-375). Springer, Cham.

Nguyen, T. M. H., Thi, T. M. N., **Ha, H. T. T.**, Truong, N. T. N., Nguyen, T. H., Cong, C. P., & Ngoc, T. M. P. (2025). Multi-stage Alzheimer's Diagnosis Using MRI Image: A Combination of Computer Vision Technique for Machine Learning Model. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 376-393). Springer, Cham.

Nguyen, X. K., Ngo, T. A., Nguyen, T., Pham, P., **Ha, H.**, & Ngo, L. (2025). A Comparative Study of Transfer Learning Techniques with BERT Family Models in Early Diagnosis of Alzheimer's Disease. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 373-382). Springer, Cham.

Luu, T., Dang-Hieu, A., Ly, T., Pham, H. T., **Ha, H.**, & Ngo, L. (2025). Developing a Photoplethysmography-Based Automated Sleep-Stage Scoring Framework Using Artificial Intelligence. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 394-408). Springer, Cham.

Thanh, N. N. T., Thanh, **H. H. T., Thanh**, H. P., Thi, L. N., Trong, N. H. T., & Hoai, T. T. T. (2025). Recent Approaches on Acoustic and Language-Based Analysis in Alzheimer's Disease: A Literature Review. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 459-491). Springer, Cham.

Huynh, V., Nguyen, Q., Do, H., Le, A., Vo, T., & **Ha, H.** (2025). An Assessment of Alpha Music Effects on Brain-Computer Interface Performance: A Preliminary Study. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 555-575). Springer, Cham.

Nguyen, M., Duong, N. V. B., Duc, V. T., Phan, C., Phan, T., & **Ha, H.** (2025). Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 408-427). Springer, Cham.

2024

Chu, L. H., Chau, C. Q., Kamel, N., **Thanh, H. H. T.**, & Yahya, N. (2024). Functional excitation-inhibition ratio for social anxiety analysis and severity assessment. Frontiers in Psychiatry, 15, 1461290.

Nguyen, H. D., Huynh, T., Nguyen, M., Nguyen, H., Bui, L. C., Ta, T. M., ... & **Ha, H.** (2024). Comparing the Hippocampal Volume Between Patients with Alzheimer's Disease and Healthy Controls: An MRI Analysis using the Harmonized Hippocampal Protocol. Brain Science Advances, 10(4), BSA-2024.

Van Tran, L., Le, A. P. H., Le, T. T. M., Nguyen, M. T. T., Nguyen, Q. L., & **Ha, H. T. T.** (2024). Comparing the efficacy of online vs. offline methods in modulating emotional responses through mental stress tasks. *Vietnam Journal of Science, Technology and Engineering*, *66*(2), 107-113.

Nguyen, M., Pham, B., Vo, T., & **Ha, H.** (2024). Revealing the heterogeneity of plasma protein and cognitive decline trajectory among Mild Cognitive Impairment patients by clustering of brain atrophy features. Brain Multiphysics, 100093.

Trong, K. N., Luong, N. N., Tan, H., Trung, D. T., Thanh, **H. H. T.**, & Thanh, B. N. (2024). Real-time Single-Channel EOG removal based on Empirical Mode Decomposition. EAI Endorsed Transactions on Industrial Networks and Intelligent Systems, 11(2).

2023

Nguyen, K., Nguyen, M., Dang, K., Pham, B., Huynh, V., Vo, T., ... & **Ha, H.** (2023). Early Alzheimer? s disease diagnosis using an XG-Boost model applied to MRI images. *Biomedical Research and Therapy*, *10*(9), 5896-5911.

Duong, H., Nguyen, T., Pham, D., Duong, K., Mieu, L., Nguyen, H., ... **Ha, H. T. T.** & Ngo, L. (2023, December). HosRev: A Comprehensive Aspect-Category Sentiment Analysis Dataset on Hospital Reviews in Vietnam. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-6). IEEE.

Do, T. T., Van Tran, L., Le, T. A., Le, T. M. T., Duong, L. A. H., Nguyen, T. H., ... & **Ha, H. T. T.** (2023, December). Stress Prediction Using Machine-Learning Techniques on Physiological Signals. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-7). IEEE.

Luu, T., Luu, T., Bui, T. T., Trieu, C., Truong, D., Hoang, N., ... & **Ha, H.** (2023, December). VHCor: Vietnamese Healthcare Corpus-A Comprehensive Dataset for Vietnamese Medical Department Recognition. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-6). IEEE.

Cao, T. H. M., Le, A. P. H., Tran, T. T., Huynh, V. K., Pham, B. H., Le, T. M., ... & **Ha, H. T. T.** (2023). Plasma cell-free RNA profiling of Vietnamese Alzheimer's patients reveals a linkage with chronic inflammation and apoptosis: a pilot study. Frontiers in Molecular Neuroscience, 16.

Phan, T. N. M., Nguyen, H. T., Huynh, T. N. M., Nguyen, T. H., Tran, T. N. Y., & **Ha, H. T. T.** (2022, December). Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 719-737). Cham: Springer Nature Switzerland.

Le, A. P. H., Nguyen, Q. L., Pham, B. H., Cao, T. H. M., Van Vo, T., Huynh, K., & **Ha, H. T. T.** (2023). SALAD: Syringe-based Arduino-operated Low-cost Antibody Dispenser. HardwareX, e00455.

Nguyen, T. T. Q., Hoang, C. B. D., Hoang Le, M. D., Anh Vo, N. T., Quang, H., Nguyen, C. M., ... & **Thanh Huong Ha, T.** (2023). Assessing cognitive decline in Vietnamese older adults using the Montreal Cognitive Assessment-Basic (MoCA-B) and Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) during the COVID-19 pandemic: a feasibility study. *The Clinical Neuropsychologist*, 1-19.

Nguyen, M. T. D., Xuan, N. Y. P., Pham, B. M., Do, H. T. M., Phan, T. N. M., Nguyen, Q. T. T., ... & **Ha, H. T. T.** (2023). Optimize temporal configuration for motor imagery-based multiclass performance and its relationship with subject-specific frequency. *Informatics in Medicine Unlocked*, *36*, 101141.

Do, H., Hoang, H., Nguyen, N., An, A., Chau, H., Khuu, Q., ... & **Ha, H.** (2023). Intermediate effects of mindfulness practice on the brain activity of college students: An EEG study. IBRO Neuroscience Reports, 14, 308-319.

2022

An, A., Hoang, H., Trang, L., Vo, Q., Tran, L., Le, T., ... & **Ha, H.** (2022). Investigating the effect of Mindfulness-Based Stress Reduction on stress level and brain activity of college students. *IBRO neuroscience reports*, *12*, 399-410.

Dang, K., Vo, T., Ngo, L., & **Ha, H.** (2022). A deep learning framework integrating MRI image preprocessing methods for brain tumor segmentation and classification. IBRO neuroscience reports, 13, 523-532.

Nguyen, L. Q., An, H. M. A., Tran, V. Q. T., Le, P. H. A., Le, M. T., Nguyen, T. K., & **Ha, T. T. H.** (2022). Investigation of the effect of alpha wave-containing music on the cognitive performance and brain activity of university students. *Vietnam Journal of Science, Technology and Engineering*, *64*(3), 69-75.

Nguyen, D., Nguyen, H., Ong, H., Le, H., **Ha, H.**, Duc, N. T., & Ngo, H. T. (2022). Ensemble learning using traditional machine learning and deep neural network for diagnosis of Alzheimer's disease. *IBRO Neuroscience Reports*, *13*, 255-263.

Ngan Thanh, L., Anh Hoang Lan, D., Dung Xuan, N., Khiet Thi Thu, D., Chau Nu Ngoc, P., Thuong Hoai, N., & **Huong, H. T. T.** (2022). Classification of Four-Class Motor-Imagery Data for Brain-Computer Interfaces. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 763-778). Springer International Publishing.

Do, T. T., Nguyen, T. H., Le, T. A., Nguyen, S. A. T., Nguyen, Q. T. N., Tran, T. Q. V., ... & **Ha, H. T. T.** (2022). Automated EOG removal from EEG signal using independent component analysis and machine learning algorithms. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 1001-1016). Springer International Publishing.

Van Toi, V., Nguyen, T. H., Long, V. B., & **Huong, H. T. T.** (2022). 8th International Conference on the Development of Biomedical Engineering in Vietnam.

Ong, H., Le, H., Nguyen, H., Nguyen, D., **Ha, H.**, Ngo, H. T., & Duc, N. T. (2022). A Machine Learning Framework Based on Extreme Gradient Boosting for Intelligent Alzheimer's Disease Diagnosis Using Structure MRI. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 815-827). Springer International Publishing.

Anh Hoang Minh, A., Nghia Trung, N., Thuong Hoai, N., Khiet Thi Thu, D., & **Ha, H. T. T.** (2022). The Effect of Mindfulness-Based Stress Reduction Among Various Subject Groups: A Literature Review. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 869-899). Springer International Publishing.

Le, A. P. H., Tran, T. T., Cao, T. H. M., Le, T. M., Le, P. T., & **Huong, H. T. T.** (2022). Evaluate and Optimize Cell-Free RNA Extraction Methods to Apply for Alzheimer's Disease Biomarkers Detection. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 591-609). Springer International Publishing.

Le, A. P. H., & **Huong, H. T. T.** (2022). The Role of Cell-Free RNA in Clinical Diagnosis and Evaluation of Cell-Free RNA Extraction Methods. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 637-656). Springer International Publishing.

Nguyen, T. D., & **Ha, T. T. H.** (2022). Bipolar disorder traits: An electroencephalogram systematic review. *Vietnam Journal of Science, Technology and Engineering*, *64*(4), 84-90.

2021

Thinh, T. V. Q., Quang, N. L., Anh, A. H. M., Long, T. H., Hieu, N. T., Van Luan, T., ... & **Huong, H. T. T.** (2021, May). Effect of Alpha Wave-Containing Music on the Psychological Stress Level of Hospital Staffs and their Cognitive Performance. In 2021 IEEE 3rd Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS) (pp. 25-28). IEEE.

2020

Fujita, H., ... & **Ha, H.** (2020, September). SEED: A Framework for Stress Estimation Using Emotiv Devices. In *Knowledge Innovation Through Intelligent Software Methodologies, Tools and Techniques: Proceedings of the 19th International Conference on New Trends in Intelligent Software Methodologies, Tools and Techniques (SoMeT_20)* (Vol. 327, p. 435). IOS Press.

Pham, V. M., **Ha, H. T.**, & Thakor, N. (2020). Microfluidic Culture Platforms in Neuroscience Research. *Handbook of Neuroengineering*, 1-39.

Tran, N. M. P., Huynh, T. L. D., Phan, B. N., Dang, N. N. T., Phan, T. B., **Ha, H. T. T.**, ... & Nguyen, T. H. (2020). Conjugated linoleic acid grafting improved hemocompatibility of the polycaprolactone electrospun membrane. *International Journal of Polymer Science*, *2020*, 1-8.

Tran, N., Le, A., Ho, M., Dang, N., **Thi Thanh, H. H.**, Truong, L., ... & Hiep, N. T. (2020). Polyurethane/polycaprolactone membrane grafted with conjugated linoleic acid for artificial vascular graft application. *Science and Technology of Advanced Materials*, *21*(1), 56-66.

2018

Ha, H. T., Leal-Ortiz, S., Lalwani, K., Kiyonaka, S., Hamachi, I., Mysore, S. P., ... & Kim, S. A. (2018). Shank and zinc mediate an AMPA receptor subunit switch in developing neurons. Frontiers in molecular neuroscience, 11, 405.

2016

Ha, H., & Huguenard, J. (2016). Criminal Minds: Cav3. 2 Channels Are the Culprits, but NMDAR Are the Co-Conspirators. *Epilepsy Currents*, *16*(1), 36-38.

Laboratory products

BrainTrain
https://www.braintrainhcmiu.com/
https://www.facebook.com/profile.php?id=100093556011720&_rdr

BrainConnects 2023
https://brainconnects2023.wixsite.com/brain-connects
https://www.facebook.com/BrainConnects/?_rdr

BrainAnalytics
https://brainanalyticsorg.wixsite.com/brain-analytics

The Brainiac
https://www.facebook.com/BMEBrainiacs

Presentations and Posters

Talk – 2025

International

- UK - ASEAN Workshop on Partnership in AI-Powered Health Smart Technologies, Hanoi, Vietnam
- Meeting with J&J leaders, Vietnam

Domestic

- 12th Scientific Conference on diagnostic imaging 2025, Ho Chi Minh City, Vietnam

Talk – 2024

International

- Biomarkers - FORUM: Bridging Neuroscience Discoveries Towards Better Therapeutic Outcomes - Technology in Dementia, 18th International Congress of the Asian Society Against Dementia 2024, Malaysia.
- Intervention Using Cognitive Apps for Patients with Mild Cognitive Impairment (MCI) - Technology in Dementia, 18th International Congress of the Asian Society Against Dementia 2024, Malaysia. (https://asad2024.com/programme-detailed)
- AI Connect II — AI Frontiers: Bolstering Scientific Exchange and Research, Vietnam.
- Webinar IBRO-EEC edition: Latest Advancements in Understanding Alzheimer's Disease - Global Neuroscience Horizons.
(https://www.youtube.com/watch?v=k_JSWhFdxF4)
(https://drive.google.com/file/d/1bBIzLDrI9WXTFPMFu_5PjLf5k1k83y3A/view?usp=sharing)
- Academic visit to establish a research collaboration that supports comparative studies on living with dementia - University of Hertfordshire.

Domestic

- “Mathematics in Biomedicine”, Math Open Day MOD 2024
- “Reaching for Greater Heights” Talkshow, Auditorium B, Gia Dinh People's Hospital, Vietnam. (https://www.facebook.com/share/p/17hpsziHfz/)
- Scientific research methods course, The Youth Scientific and Technological Promotion Center (TST). (https://www.facebook.com/share/p/18yfKDdRJG/)
- Seminar “Students and scientific research”, Vietnam.
- Seminar “Sharing experiences as a PhD student”, Vietnam.
- Report “Sharing experiences in appointing Professors and Associate Professors in some countries”, Vietnam.

Talk – 2023

International

- Lab meeting - Nanyang Technological University (NTU)
- Neuroengineering approaches for stress detection and intervention in low-middle-income countries, 13th Mind-Body Interface International Symposium
- Rencontres du Vietnam: The first International Symposium of Nano Life Science: Nano Biotechnology, Biosensor, Computation (NanoBioCoM2023), Quy Nhon, Vietnam
- Webinar “Crafting a Successful IBRO Grant Application” (https://www.youtube.com/watch?v=-IVldyvVPQI&t=5s) (https://drive.google.com/file/d/1-cbNY6orREuDcSSu_kkGBLBQyuBb1K_q/view?usp=sharing)
- Aspects Surrounding Quality of Life, Hyper Interdisciplinary Conference in Vietnam 2023: Advancing Quality of Life: Bridging Human Innovation (https://global.lne.st/news/ph/2022/12/17/hicvn2023keynote/)

Domestic

- RMIT's Psychology Career Workshop
- Neuroengineering Approaches for Stress Detection and Intervention in Low-Middle-Income Countries, Signal Processing and Data Science for Smart Health Conference & VISHC Summer School 2023 (VinUniversity) (https://smarthealth.vinuni.edu.vn/event/the-first-vishc-summer-school-2023-to-excel-in-research-activities-for-vinuni-undergraduates/)
- Psychology Research Seminar - RMIT University

Talk – 2022

International

- Women in Neuroscience Symposium (https://drive.google.com/file/d/1S8cSfFXgwgkdQuN9GgX3DLwhvEqsd9kV/view?usp=sharing)
- Application of AI and blockchain in medical and education, International Health Tech Conference.

Domestic

- Multi-Disciplinary Characteristics in Neurology Problems, Math Open Day MOD 2022. (https://www.facebook.com/watch/live/?ref=watch_permalink&v=1476673259500472)
- Workshop “Gen Z and Scientific research”, International University, Vietnam. (https://hcmiu.edu.vn/workshop-chuyen-de-gen-z-nghien-cuu-khoa-hoc)

Talk – 2021

International

- Effect of alpha wave-containing music on the psychological stress level of hospital staffs and their cognitive performance - The 3rd IEEE Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (Best Paper Award).
- Investigating brain and cardiovascular responses to develop a machine learning model for acute and chronic stress detection - International Biomedical Science and Engineering Symposium, Gwangju Institute of Science and Technology, November 25-26, 2021.
- World Dementia Council (https://drive.google.com/file/d/1fd5GjcC9olmurbcMrrPptxZlRWWnRh-h/view?usp=sharing)

Domestic

- Workshop “Finding creative ideas about scientific research”, The Youth Scientific and Technological Promotion Center (TST). (https://khoahoctre.com.vn/workshop-tim-kiem-y-tuong-sang-tao-ve-nghien-cuu-khoa-hoc-2/)
- The Future is Female - Fulbright University Vietnam Career Day 2021
- Guest lecture: Research methodology and application of neuroscience - Vietnam Biology Olympiads team, High School for the Gifted – VNU-HCM
- Characteristics and steps to conduct scientific research - Ho Chi Minh Communist Youth Union, Vietnam
- Seminar “Women in Innovation and Creativity”. (https://www.dropbox.com/scl/fi/lm87grzx6hbz38fiqqeah/Huong_Women-in-Innovation-Panelist.jpeg?rlkey=1dhmdo0j0evbom6ed8q5groj6&st=sgiwl67t&dl=0)

Talk – 2020

International

- SEED: A Framework for Stress Estimation Using Emotiv Devices, The 19th International Conference on New Trends in Intelligent Software Methodologies, Tools and Techniques, SoMeT 2020, Kitakyushu, Japan, September 22-24, 2020.
- Brain activity associated with examination-induced stress in college students, The 2nd University of Tsukuba-University of Science Joint Young Researcher Meeting in Biomedical Sciences, The 1st Student Conference, University of Science.

Domestic

- Application of Artificial Intelligence to develop tools for diagnosing Alzheimer’s Disease, The 2nd Symposium on Alzheimer’s Disease at University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam.
- Towards a non-invasive diagnosis for Alzheimer’s Disease, The 1st Symposium of the Vietnam Alzheimer Network, National Geriatric Insitute, Ho Chi Minh city, Vietnam.
- Lecture: A research methodology course, I-IVY education, Ho Chi Minh, Vietnam

Talk – 2019

International

- Perspective: Brain Aging in Vietnam: Society needs and research approach, The 9th Neuroscience International Symposium: Bridging Brains – Enriching Minds 2019 in the Philippines

Domestic

- Introduction into Mindfulness Based Stress Reduction, International University, Vietnam National University, Ho Chi Minh City, Vietnam.
- Women in Science, Unfold the V, Ho Chi Minh City, Vietnam.
- The common myths and the neurobiology of autism, Hoa Sen University, Ho Chi Minh City, Vietnam.
- Developing diagnostic methods for neurodegenerative diseases, Symposium on Laboratory Test and Quality Control, University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam.
- Workshop “Working with scientific posters” (https://www.facebook.com/share/p/1DSo6E7mgg/)

Talk – 2018

- EEG applications in Alzheimer’s Disease Detection, Alzheimer’s Disease Awareness Day Symposium, University of Medicine and Pharmacy Medical Center, Ho Chi Minh City, Vietnam

Poster – 2023

- Comparing Hippocampus Volume in Alzheimer’s Disease and Healthy Controls: A MRI Analysis with HarP T
- Analyzing the effect of stimulation frequency on the performance of Machine Learning Classification on public Migraine EEG dataset
- Plasma cell-free RNA Profiling of Vietnamese Alzheimer’s patients: A pilot study
- The Role of Quantitative EEG in Assessment of Cognitive Changes after Ischemic Stroke
- Evaluating stress responses in Vietnamese university students to Trier Social Stress Tasks using multiple measures as a reference for a Point-of-care stress diagnostic device

Poster – 2022

- Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly

Poster – 2021

- Current state of Alzheimer's diagnosis in Vietnam and doctors' view of an Alzheimer's AI detector
- Immediate effects of Mindfulness-Based Stress Reduction
- Early diagnosis of Alzheimer's disease using Extreme Gradient Boosting (XG-Boost) on MRI images
- Development of a Cortisol-quantification test strip for stress evaluation using low-cost materials
- Mindfulness-based Stress Reduction program for college students: Effect on stress level and EEG power
- Investigating brain and cardiovascular responses to develop a machine learning model for acute and chronic stress detection

Poster – 2018

- Shank and zinc mediate an AMPA receptor subunit switch in developing neurons
  The Cold Spring Harbor Conference in Osaka, Japan
  The Molecular and Cellular Neurobiology Gordon Conference in Hong Kong.
  The Bay Area Molecular and Cellular Neuroscience meeting, California, USA.

C. Teaching experience

Current courses:

Undergraduate program

1. Brain Computer Interface (BCI) - 1 class/term & 1 term/year - 4 credits (Technical Elective course)
2. Principles of Neuro-engineering - 1 class/term & 1 term/year - 4 credits (Technical Elective course)
3. Bioethics - 1 class/term & 2 terms/year - 3 credits (Foundation course)
4. Project 1B: in vitro studies - 2 classes/term & 2 terms/year - 1 credit (Foundation course)

Graduate program

1. Research Methodology - 1 class/term & 2 terms/year - 3 credits
2. Progress in Biomedical Engineering - 1 class/term & 2 terms/year - 3 credits
3. Engineering Challenges in Medicine - 1 class/term & 2 terms/year - 3 credits
4. Advanced Biosignal Processing - 1 class/term & 2 terms/year - 2 credits
5. Computational Neuroscience and Applications - 1 class/term & 1 term/year - 2 credits
6. Biostatistics - 1 class/term & 1 term/year - 3 credits

Previous teaching experience:

(1) Coach at Research and Design workshop, Stanford D-School	2014
(2) Lead Instructor at Stanford School of Medicine Health Matters, Brain Day and RISE programs	2013-2014
(3) Teaching Assistant at Stanford Intensive Neuroscience Course	2012
(4) Teaching Assistant in the General Molecular Biology Techniques at University of Sciences- Vietnam	2011

D. Positions and Awards

Positions and Employment

Educational Quality Assurance Coordinator, School of Biomedical Engineering, International University, HCMC VNU	2021 – present
Chair, Department of Tissue Engineering and Biomedical Engineering, School of Biomedical Engineering, International University, HCMC VNU	2020 – present
Research coordinator, School of Biomedical Engineering, International University, HCMC VNU	2019 – 2020
Lecturer, School of Biomedical Engineering, International University, HCMC VNU	2018 – present
Psychiatry Department, School of Medicine, Stanford University	2014 – 2015
Department of Neurology and Neurological Sciences, School of Medicine, Stanford University	2012 – 2014
Research assistant at the Oxford University Clinical Research Unit (OUCRU)	2011 – 2012

Other Experience and Professional Memberships

The Rising Talents List of the Precision Nanomedicine Editorial Board	2024 – present
Member of IBRO Early Career Committee	2024 – present
Member of the National Academies' Workshop Planning Committee on Engaging Scientists in Shared Responsible Innovation in Neuroscience in Southeast Asia	2023 – present
Member of editorial board, Brain Behavior and Immunity Integrative	2023 – present
Guest Editor, IBRO Neuroscience Report Mini-series for the IBRO APRC Alumni	2021 – present
Member, Brain connects (international organization)	2019 – present
Organizing Committee Member, BrainConnect (which includes supporting the Special Research Topic on Frontier Human Neuroscience)	2019 – present
Vice chair, Bioethics committee, School of BME, International University, HCMC VNU	2019 – 2021
Advisory board, International School at Park City Hanoi (ISPH), Vietnam	2018 – present
Member, Biomedical Engineering (BME) society, Vietnam	2018 – present
Student Program Committee Member, Stanford Neurosciences Graduate Program	2015 – 2018
Reviewer (Journal of Neuroscience, Nature Communications, Neuroscience, Neuron, Vietnam Journal of Science, PNAS)	2015 – present
Leader, Vietnam Book Drive project, VEFFA	2013 – 2014
Executive Member, Vietnam Education Foundation Fellow Association (VEFFA)	2013 - 2014
Managing Editor and Founder, Vietnam Journal of Science	2013 – 2017

Individual Awards

Second Prize of the 5th Vietnam Medical Achievement Award, Vietnam.	2025
Vietnamese Women	2024
Advanced Youth of Ho Chi Minh City following Uncle Ho's teachings	2024
Outstanding achievements in studying and following Ho Chi Minh's ideology, morality and style in the period 2021-2024	2024
Outstanding female intellectual of Ho Chi Minh City 2019-2024, Vietnam	2024
Golden Globe Science and Technology Award, Vietnam	2023
National Outstanding Teacher	2023
Outstanding Young Citizen of Ho Chi Minh City	2023
Women of the Future Awards Southeast Asia	2023
Vietnamese Outstanding Young Face Award	2023
Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada	2023
Second Prize, China-ASEAN Innovation and Entrepreneurship Competition	2023
International Awards L'Oréal-UNESCO for Women in Science	2022
Top 4 Social Business Creation, HEC Canada	2022
Third Prize, National Technological Creation Contest, Vietnam	2022
Second Prize, Ho Chi Minh City Creative Awards	2021
SUN* Award, Tech Planter Demo day in Vietnam	2021
Top 5 Business Innovation Idea, Canada	2021
Staff Excellent Award, International University, VNU, Ho Chi Minh City, Vietnam	2021
Young Talent Award, Vietnam National Youth Association	2020
Top 5, BAMBUuP Disruption in Health Care, Vietnam	2020
Top 8, Ho Chi Minh City Artificial Intelligence Competition, Vietnam	2020
Early Career Award, International Brain Research Organization	2020
ALBA-FKNE-YIBRO Diversity Grant for participating in the FENS 2020 Virtual Forum	2020
Travel Grant to Attend Science Conference (Stanford University Office of Graduate Education)	2018
Faculty for the Future Fellowship (The Schlumberger Foundation, Paris, France)	2015 - 2018
Outstanding Service Award (The Vietnam Education Foundation Fellow Association - VEFFA, US)	2014
Neurobiology Course Summer Scholarship (The Marine Biological Laboratory, Massachusetts, US)	2014
Vietnam Education Foundation Fellowship (The Vietnam Education Foundation, Virginia, US)	2012 – 2014
Stanford Graduate Fellowship (Stanford University, California, US)	2012 – 2015
Five Goods Award (The People Committee, Ho Chi Minh City, Vietnam)	2012
Eureka Scientific Award (The Center of Technology and Development, Ho Chi Minh City, Vietnam)	2011
Award for Exceptional Graduation (The University of Science, Ho Chi Minh City, Vietnam)	2011
Scholarships for outstanding high school and college students (The Odon Vallet Foundation, Sumitomo Corporation, and Lawrence S. Ting Memorial Funds).	2006 – 2011

Awards from my lab students

(2024) AmCham Scholarship Program.

(2024) Pony Chung Scholarship.
(2024) Vingroup Scholarship – VinUniversity.
(2024) Travel award NEURO2024 Conference.
(2024) IBRO-APRC Associate School on “Neuro-Enhancement for Brain Health”.
(2024) First Prize, Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment.”
(2024) Silver Medal in the “Design, manufacture, application” award, Vietnam.
(2023) IUPAB Travel Award, Rencontres du Vietnam: The First International Symposium of Nano Life Science: Nano Biotechnology, Biosensor, Computation (NanoBioCom2023), Vietnam.
(2023) 5-min Poster Blitz, 13th Mind-Body Interface International Symposium.
(2023) Travel award, 13th Mind-Body Interface Symposium, Taiwan.
(2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.
(2023) Second Prize, China-ASEAN Innovation and Entrepreneurship Competition.
(2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.
(2023) Third prize in the contest "Digital Transformation - Creativity Challenge", Ho Chi Minh City, Vietnam.
(2023) ADAI Fellowship, Ho Chi Minh City, Vietnam.
(2022) UNLEASH HACKS: Mental Health Europe.
(2022) Top 4 Social Business Creation, HEC Canada.
(2022) Third Prize, National Technological Creation Contest, Vietnam.
(2022) ADAI Fellowship, Ho Chi Minh City, Vietnam.
(2022) ITR Graduate Fellowship, Ho Chi Minh City, Vietnam.
(2021) Second Prize, Ho Chi Minh City Creative Awards.
(2021) SUN* Award, Tech Planter Demo day in Vietnam.
(2021) Top 5 Business Innovation Idea, Canada.
(2020) Top 5, BAMBUuP Disruption in Health Care, Vietnam.
(2020) Top 8, Ho Chi Minh City Artificial Intelligence Competition, Vietnam.

E. Funding

No.	*Title of project*	*ID number & Level of management*	*Duration*	*Cost (EUR)*	*Role*	*Achieved Outcome*
1	An ultra-sensitive plasma p-tau 217 immunosensor using screen-printed carbon electrodes modified with MWCNTs/metallic nanoparticles and oriented immobilized antibody for Alzheimer’s Disease detection		2024-2025		Principal investigator	
2	Multimodal Screening & Management System of Mental Wellbeing in Low Resource Settings		2023-2024		Co-principal investigator	
3	BrainConnects 2023 Conference Grant		2023	6000	Host	
4	Investigation and Implementation of Solutions for Challenges in	Medical Research Council,	2020-2021	25000/ 1 year	Main investigator	1 international conference

	Mental Issues of the Elderly in Vietnam by Networking with Advanced Countries	United Kingdom				
5	Toward a non-invasive blood test for Alzheimer Disease	Vietnam Alzheimer Network	2020–2022	10000/ 2 years	Principal investigator	01 international conference publication
6	Grant for Outstanding Research Group	#NCM2020-28-01 HCMC VNU	2020 - 2025	3800/ year	Main investigator	01 international conference publication 01 international conference talk
7	Development and Evaluation of a Cortisol quantification test strip for reliable and low-cost stress diagnosis.	C2022-28-01	2021 - 2023	3000	Principal investigator	01 Q1-paper publication 01 international conference publication
8	Investigation on establishing a large database of EEG and video recordings of Vietnamese people in application for intelligent control and primary motor rehabilitation in epileptic patients	# KC-4.0-07/19-25, Ministry of Science and Technology	2020-2023	19000	Main investigator	01 international conference publication
9	Investigating a model of Tele-EEG coupled with ultrasound function emitter on stress evaluation of college students	# KC-4.0-07/19-25, HCMC VNU	2020-2022	3800	Main investigator	01 international conference publication 01 international conference talk
10	Artificial Intelligent system for automatic diagnosis of Alzheimer's disease using neuroimaging	#T2019-05-BME, Internal grant of International University, HCMC VNU	2019 - 2021	1900	Main investigator	01 international conference publication
11	Investigate the effects of mindfulness-based stress reduction (MBSR) program on stress level and brain activity of college students	#T2019-04-BME Internal grant of International University, HCMC VNU	2019-2021	1900	Principal investigator	01 international conference publication 01 international conference talk
12	Solid-state patch clamp platform to diagnose	#5R33MH100717-04, National	2013-2018		Student Participant	01 Q2-paper publication

	autism and screen for effective drug	Institute of Health, US				
13	Astrocytic Control of GABA Inhibition in Epilepsy	#5R01NS090 911-04, National Institute of Health, US	2016-2018		Student Participant	01 Q1-paper publication

E. REFERENCE

Toi V. Van, Ph.D.
School of Biomedical Engineering, International University
vvtoi@hcmiu.edu.vn

John Huguenard, Ph.D.
Stanford University School of Medicine
john.huguenard@stanford.edu

Sally A. Kim, Ph.D.
Amherst University
sakim@amherst.edu

Craig Garner, Ph.D.
German Center for Neurodegenerative Diseases
Craig-Curtis.Garner@dzne.de

Hoan T. Ngo, Ph.D. Website Google Scholar

OFFICE Weinstein Computer Science 116
Florida Southern College
Lakeland, FL 33801 USA

E-mail: hngo@flsouthern.edu
Total number of citations: > 1,400
Citation h-index: 19

RESEARCH INTERESTS

Artificial Intelligence for Health, Cancer and Neurodegenerative Diseases Diagnostics, Digital Health, Computational Bioengineering, Diagnostics in Low-Resource Settings

TRAINING

Johns Hopkins University, Baltimore, Maryland USA May, 2021 - July, 2023

Postdoctoral Scholar
Advisor: Jeff "Tza-Huei" Wang, Ph.D.

Duke Eye Center, Durham, North Carolina USA April, 2017 - December, 2017

Postdoctoral Scholar
Advisor: Joseph Izatt, Ph.D. and Cynthia Toth, MD

EDUCATION

Duke University, Durham, North Carolina USA August, 2011 - March, 2017

Ph.D. in Biomedical Engineering, May, 2017
Advisor: Tuan Vo-Dinh, Ph.D.

Korea Maritime University, Busan, South Korea August, 2006 - August, 2008

M.E, Control and Instrument Engineering, August, 2008

University of Technology, Ho Chi Minh City, Vietnam September, 2001 – April, 2006

B.E, Mechatronics, April, 2006

RESEARCH GRANTS

- August, 2024 - July, 2027: Adapting a machine learning algorithm to predict thyroid cytopathology in LMIC. Sponsor: National Institutes of Health (National Cancer Institute). Award number: # 4R33CA268428-03 (3 years, $287,551/year). Co-PI.
- January, 2025 - January, 2028: The application of Artificial Intelligence (AI) in the diagnosis and prognosis of Alzheimer's disease. Sponsor: Vietnamese Ministry of Science and Technology. Program KC.10/21-30 (approx. $284,437). Co-I.

SELECTED PUBLICATIONS

Below are 5 selected publications. For full list, refer to my Google Scholar

1. Aiden Coffey, Vladimir Cuc, **Hoan Thanh Ngo**, Walter Lee. *Thyroid Cancer Detection using Smartphone-captured Cytopathology Images and Foundation*

Model (Accepted for Medical Imaging with Deep Learning (MIDL), Salt Lake City, Utah, United States, Jul 09 2025).

2. Quan Thanh Huynh, Hieu Xuan Le, Lien Tran, Trung Nguyen, Luan Nguyen, Alex Mremi, Daniel Mbwambo, Walter Lee, **Hoan Thanh Ngo**. *Explainable AI for Thyroid Cancer Detection using Smartphone-captured Cytopathology Images and Multi-Instance Learning* (Accepted for and presented at BMES 2024 Annual Meeting Oct 23-26, Baltimore, MD USA).
3. **Hoan T Ngo**, Patarajarin Akarapipad, Pei-Wei Lee, Joon Soo Park, Fan-En Chen, Alexander Y Trick, Tza-Huei Wang, Kuangwen Hsieh. *Rapid and portable quantification of HIV RNA via a smartphone-enabled digital CRISPR device and deep learning*. Sensors and Actuators Reports, 8, 100212, 2024.
4. Dong Nguyen, Hoang Nguyen, Hong Ong, Hoang Le, Huong Ha, Nguyen Thanh Duc, **Hoan T Ngo**. *Ensemble learning using traditional machine learning and deep neural network for diagnosis of Alzheimer's Disease*. IBRO Neuroscience Reports, 13, 255-263, 2022.
5. Quan Thanh Huynh, Phuc Hoang Nguyen, Hieu Xuan Le, Lua Thi Ngo, Nhu-Thuy Trinh, Mai Thi-Thanh Tran, Hoan Tam Nguyen, Nga Thi Vu, Anh Tam Nguyen, Kazuma Suda, Kazuhiro Tsuji, Tsuyoshi Ishii, Trung Xuan Ngo, and **Hoan T Ngo**. *Automatic acne object detection and acne severity grading using smartphone images and artificial intelligence*. Diagnostics, 12(8), 1879, 2022.

Book/Book Chapter Publications

1. **Hoan T Ngo**, Tuan Vo-Dinh, "SERS biosensors for Infectious Disease Diagnosis". Chapter 3 in the Book SERS for Point-of-care and Clinical Applications. Elsevier Science, 2022.
2. Van Toi, V., Le, T.Q., **Ngo, H.T.**, Nguyen, T.-H. (Eds.), Book "7th International Conference on the Development of Biomedical Engineering in Vietnam (BME7)", IFMBE Proceedings, Springer Singapore publisher, 2019.
3. **HT Ngo**, N Gandra, AM Fales, SM Taylor, T Vo-Dinh, "Sensitive DNA Detection and SNP Identification Using Ultrabright SERS Nanorattles and Magnetic Beads for In Vitro Diagnostics", Chapter 27 in the Book Nanotechnology in Biology and Medicine: Methods, Devices, and Applications, Second Edition, Taylor and Francis publisher, 2017.
4. T. Vo-Dinh, HN Wang, **HT Ngo**, "SERS Molecular Sentinel Nanoprobes to Detect Biomarkers for Medical Diagnostics", Chapter 24 in the Book Biomedical Photonics Handbook, Second Edition: Therapeutics and Advanced Biophotonics, Taylor and Francis publisher, 2015.

Patents

1. T Vo-Dinh, N Gandra and **HT Ngo**, "Nanoprobe compositions and methods of use thereof", US Patent 10,876,150, 2020.

Employment

Assistant Professor August, 2023 - present
Department of Computer Science & Engineering
Florida Southern College, Lakeland, Florida USA

- Developing an AI-powered system for thyroid cancer detection in low-resource settings (in collaboration with Walter Lee, MD at Duke University School of Medicine and medical doctors from Tanzania and Vietnam).
- Developing an AI-powered system for early Alzheimer's disease detection using retinal images and brain MRI images (in collaboration with medical doctors in Vietnam).
- Developing an AI-powered system for psoriasis severity indexing using smartphone images (in collaboration with Scott Elman, MD at University of Miami Miller School of Medicine).

Postdoctoral Scholar May, 2021 - July, 2023
Department of Biomedical Engineering and Mechanical Engineering
Johns Hopkins University, Baltimore, Maryland USA
Advisor: Dr. Jeff "Tza-huei" Wang

- Developed a PCR-based method and portable device for HIV viral load quantification at the point-of-care.
- Developed a CRISPR-based method and a smartphone-based device + Deep Learning for rapid and portable HIV RNA quantification.

Faculty & Vice Chair January, 2018 - April, 2021
Department of Biomedical Engineering
International University - Vietnam National University, HCM City, Vietnam

- Developed various low-cost medical devices for developing countries including a low-cost PCR machine, a smartphone-based fundus camera, a smart blind stick, etc.
- Developed AI algorithms for various medical image analysis.

Postdoctoral Scholar April, 2017 - December, 2017
Duke Eye Center
Duke University, Durham, North Carolina USA
Advisor: Dr. Joseph Izatt and Cynthia Toth, MD

- Developed an optical imaging system based on Optical Coherent Tomography (OCT) for guidance of ophthalmic microsurgeries.
- Developed algorithms for analysis of OCT angiography images.

Research Assistant August, 2011 - March, 2017
Department of Biomedical Engineering
Duke University, Durham, North Carolina USA
Advisor: Dr. Tuan Vo-Dinh

- Developed biosensors and integrated systems for point-of-care diagnostics of respiratory infection, dengue, and malaria.
- Developed biosensors and integrated systems for point-of-care diagnostics of head and neck cancer.

Founding Faculty October, 2008 - July, 2011
Department of Biomedical Engineering & School of Electrical Engineering
International University - Vietnam National University, HCM City, Vietnam

- Developed inductive plethysmography sensor for respiratory monitoring.
- Developed an algorithm for extracting fetal ECG from materal ECG.

Teaching Experience

Florida Southern College, Lakeland, Florida USA August, 2023 - present
Department of Computer Science and Engineering
Courses taught:

- Introduction to Computer Science (Python Programming)
- Computer Organization and Architecture
- Computer Science Research I & II (topic: AI in Medicine)
- Microcontroller Programming
- Introduction to Embedded Systems

International Uni. - Vietnam National Uni., HCM City, Vietnam January, 2018 - April, 2021
Department of Biomedical Engineering
Courses taught:

- Medical Imaging
- Medical Instrumentation
- Medical Instrumentation Design
- Biosignal Processing
- Biomedical Image Processing
- Artificial Intelligence for Biomedical Engineers

Duke University, Durham, North Carolina USA August, 2011 - March, 2017
Department of Biomedical Engineering
TAed:

- Biomedical Electronic Measurements I
- Advances in Photonics

International Uni. - Vietnam National Uni., HCM City, Vietnam October, 2008 - July, 2011
Department of Biomedical Engineering & **School of Electrical Engineering**
Courses taught:

- Principles of Electrical Engineering
- Control Systems

- Microcontroller Programming
- Biomedical Engineering Design

Industry Experience

Part-time employee August, 2006 - August, 2008
SonarTech Co., Ltd., Busan, South Korea

- Developed a control system for underwater robot.

Fellowships, Awards, and Honors

- Fitzpatrick Foundation Scholars Fellowship, 2014-2016
- Best Poster Award, Duke University Fitzpatrick Institute for Photonics Symposium, 2015
- Best Paper Award from SPIE, 2014
- Vietnam Education Foundation Fellow, 2011
- Graduated Summa Cum Laude and received Silver Medal, University of Technology, Ho Chi Minh City, Vietnam, 2006
- Champion of a robot contest organized by the University of Natural Science, HCM City, 2005
- Certificate of merit from ROBOCON Vietnam 2005, the largest robotic contest in Vietnam, 2005
- Intel Vietnam scholarship, 2002

Professional Affiliations

- Biomedical Engineering Society (BMES)
- International Society to Advance Alzheimer's Research and Treatment (ISTAART)
- The International Society of Optical Engineering (SPIE)

Professional Service

- Co-chair, Machine Learning II session, Biomedical Engineering Society (BMES) Annual Meeting, Baltimore, Maryland USA, October, 2024.
- Associate Editor, Frontiers in Lab on a Chip Technologies (May, 2023 – present).
- Manuscript Reviewer, Nature Communication, Sensors, etc.
- Co-organizer, the 3rd (2010), 7th (2018), and 8th (2020) International Conference on the Development of Biomedical Engineering in Vietnam.
- Organizer, the AI for Health workshop at International University - Vietnam National University, HCM City, Vietnam, 2019.

Computing Skills

- Completed courses and received certificate in the Deep Learning Specialization taught by AI expert, Dr. Andrew Ng.
- Completed course and received certificate in Fundamentals of Machine Learning for Healthcare (Stanford University).
- TensorFlow in Practice Specialization: completed the TensorFlow in Practice Specialization taught by Laurence Moroney, an AI Lead at Google.
- Python for Data Science and AI: completed IBM's Python for Data Science and AI
- PyTorch, MATLAB, COMSOL Multiphysics; OpticStudio by Zemax; Mathematica for symbolic mathematical computing.

Engineering and Life-science Skills

- Device Engineering Skills: Electrical and Electronics Circuit Design; Printed Circuit Board (PCB) Design; Microcontroller Programming (Arduino, ESP32, Raspberry Pi, Raspberry Pi Pico); FPGA Programming; LabVIEW; Mechanical Design and 3D Modeling (SolidWorks, Fusion 360); Machining (CNC, Milling, Lathing); Rapid Prototyping using 3D Printing and Laser Cutting, Thermoforming; Mobile App Prototyping using MIT App Inventor 2.
- Micro/Nano Fabrication and Characterization Skills: Nanochip Fabrication; Nanoparticle Synthesis; Photolithography; Electron Beam Lithography; Plasma Asher; E-beam & Thermal Metal Evaporator; DC Sputter Coating of Metals; Sputter Coater; Reactive Ion Etcher (RIE); Plasma Enhanced Chemical Vapor Deposition System (PECVD); Scanning Electron Microscope (SEM); Transmission Electron Microscope (TEM); X-Ray Photoelectron Spectrometer (XPS); Raman and surface enhanced Raman Spectroscopy using Renishaw InVia Raman Microscope and Horiba Jobin Yvon LabRam ARAMIS; UV-Vis-NIR Spectrophotometer; Atomic Force Microscope (AFM); Near-field Scanning Optical Microscope (NSOM); Fluorescence Microscope; Olympus FV1000 Multiphoton Microscope; Optical Coherent Tomography (OCT); Clean room skills.
- Life Science Skills: Cell Culture; Cell Counting; Multi-mode Plate Reader; Nucleic Acid (DNA and RNA) Extraction; Real time PCR; Digital PCR using QuantStudio chip; Isothermal nucleic acid amplification (LAMP and RPA); CRISPR Detection; DNA Probe Design; Gel Electrophoresis; Bioconjugation.

CURRICULUM VITAE

HOANG TIEN TRONG, NGHIA

Male
175 Military Hospital
Neurology Department

786 Nguyen Kiem Street, Ward 3, Go Vap Dist., Ho Chi Minh city, Vietnam

TEL. +84-696-41272
MB. + 84-976-035-999
E-Mail. dr.hnghia@gmail.com

Employment

Sep 2011 – Aug 2018	Staff in Neurology Department, 175 Military Hospital, Ho Chi Minh city, Vietnam
2019 - now	Head of Neurology Department, 175 Military Hospital, Ho Chi Minh city, Vietnam

Current position

Head of Neurology Department, 175 Military Hospital, Ho Chi Minh City, Vietnam

Member of the Executive Committee of The Vietnam Neurological Association

General Secretary of The Vietnam Association of Clinical Neurophysiology

Official representative of The International Federation of Clinical Neurophysiology and The Asian and Oceanian Chapter of the IFCN in Vietnam

Commissary of The Southeast Asian Therapeutic Plasma Exchange Consortium

Education

2004 – 2011	Military Medicine University, Hanoi, Vietnam
2013	Interns at Singapore General Hospital
2014	Interns at Seoul National University Hospital
2015	Interns at Royal North Shore Hospital, Sydney, Australia
2016	Interns at Western Pacific Office, WHO, Manila, Philippines
2016 - 2018	M.D. at University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam
2019	Fellows at Chung Buk National University Hospital, Korea

2018 - 2020	M.D.- Level II, PERIPHERAL NERVOUS SYSTEM DISORDERS at Milan University, Italy
2020 - now	Ph.D. at University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam

Publications

Publications in English

1. Hoang NTT. Guillain Barré syndrome in a Ho Chi Minh City, Vietnam hospital. Paper presented at: PNS Annual Meeting 2017; 2017July 8 -12; Barcelona, Spain.
2. Hoang NTT. Guillain-Barré Syndrome in a teaching hospital at Ho Chi Minh City, Vietnam. Paper presented at: PNS Annual Meeting 2018; 2018 Jul 22 – 25; Baltimore, Maryland, USA.
3. Hoang NTT. A case of Guillain-Barré syndrome with Treatment-Related fluctuation. Paper presented at: PNS Annual Meeting 2019; 2019 Jun 22 – 26; Genoa, Italy.
4. Hoang NTT. Multicenter study investigating the association of GBS with Flavivirus and other Arbovirus in Asia-organizational challenges. Paper presented at: PNS Annual Meeting 2019; 2019 Jun 22 – 26; Genoa, Italy.
5. Hoang NTT. Multicenter study investigating the association of GBS with Flavivirus and other Arbovirus in Asia-Patient characteristics. Paper presented at: PNS Annual Meeting 2019; 2019 Jun 22 – 26; Genoa, Italy.
6. Hoang NTT. Evaluation of treatment results in peripheral facial palsy by using clinical and nerve conduction study, Paper presented at: 13th Biennial convention of the ASEAN Neurological Association; 2019 September 20-22; Yangon, Myanmar.
7. Hoang NTT. H-reflex changes in diabetic polyneuropathy. PNS Annual Meeting 2020; 2020 June 28.
8. Viswanathan S, Hiew FL, Siritho S, Apiwattanakul M, Tan K, Quek AML, Estiasari R, Remli R, Bhaskar S, Islam BM, Aye SMM, Ohnmar O, Umapathi T, Keosodsay SS, Hoang NTT, Yeo T, Pasco PM. Impact of Covid-19 on the therapeutic plasma exchange service within the South East Asian region: Consensus recommendations and global perspectives. J Clin Apher. 2021 Dec;36(6):849-863. doi: 10.1002/jca.21937. Epub 2021 Oct 25. PMID: 34694652; PMCID: PMC8646799.
9. Hiew FL, Thit WM, Alexander M, Thirugnanam U, Siritho S, Tan K, Mya Aye SM, Ohnmar O, Estiasari R, Yassin N, Pasco PM, Keosodsay SS, Hoang NTT, Islam MB, Wong SK, Lee S, Chhabra A, Viswanathan S. Consensus recommendation on the use of therapeutic plasma exchange for adult neurological diseases in Southeast Asia from the Southeast Asia therapeutic plasma exchange consortium. J Cent Nerv Syst Dis. 2021 Nov 25;13:11795735211057314. doi: 10.1177/11795735211057314. PMID: 35173510; PMCID: PMC8842418.
10. Nguyen-Le TH, Do MD, Le LHG, Nhat QNN, Hoang NTT, Van Le T, Mai TP. Genotype-phenotype characteristics of Vietnamese patients diagnosed with Charcot-Marie-Tooth disease. Brain Behav. 2022 Sep;12(9):e2744. doi: 10.1002/brb3.2744. Epub 2022 Aug 8. PMID: 35938991; PMCID: PMC9480926.
11. Rattanathamsakul N, Siritho S, Viswanathan S, Hiew FL, Apiwattanakul M, Tan

K, Thirugnanam UN, Yeo T, Quek AML, Estiasari R, Remli R, Aye SMM, Ohnmar O, Hoang NTT, Pasco PM. Facilities, selection, outcome measurement, and limitations of therapeutic plasma exchange for neuroimmunological disorders: The South East Asian survey study. J Clin Apher. 2023 Mar 9. doi: 10.1002/jca.22047. Epub ahead of print. PMID: 36896493.

12. Phan X, Doan V, Tran T, Ha V, Tran T, Dinh T, et al. Clinicoepidemiology of Benign Paroxysmal Positional Vertigo: Initial Utilizing Videonystagmography in the Department of Neurology. 9th International Conference on the Development of Biomedical Engineering in Vietnam. BME 2022. IFMBE Proceedings, vol 95. Springer, Cham. https://doi.org/10.1007/978-3-031-44630-6_69.
13. Viswanathan S, Vijayasingham L, Laurson-Doube J, Quek A, Tan K, Yeo T, et al. Multi-actor system dynamics in access to disease-modifying treatments for multiple sclerosis in Southeast Asia: A regional survey and suggestions for improvement. Multiple Sclerosis and Related Disorders. 2024;85:105555.
14. Hoang N, Umapathi T, Duc N, Hieu N, Thao M. Genetic landscape of Charcot–Marie–Tooth disease in Vietnam: A prospective multicenter study. Journal of Neuromuscular Diseases. 2025;12.
15. Quang H, Nguyen-Martinez A, Hoang N, Mai L. The Comprehensive Neuropsychological Interview. 2025. p. 37-48.
16. Ly M, Vo N, Hoang N. AB-467. Motor-evoked potentials induced by transcranial magnetic stimulation in healthy people at Military Hospital 175. Clinical Neurophysiology. 2025;172:S63-S4.
17. Dang LH, Nhu YHT, Trong NHT, Van Anh VD. AB-520. Detection of abnormal EEG findings in patients with epilepsy using natural sleep EEG in the Clinical Neurophysiology Center at Military Hospital 175. Clinical Neurophysiology. 2025;172:S74
18. Doan V, Huynh D, Huynh T, Hoang N. AB-507. First-onset seizure in people over 50 years old: Clinical and electroencephalographic aspects in Military Hospital 175. Clinical Neurophysiology. 2025;172:S72.
19. Phan XUH, Đoan VAV, Huynh TNY, HuYnh ĐL, Hoang TTN. AB-635. Prevalence and classification of the horizontal semicircular canal benign paroxysmal positional vertigo in Vietnam. Clinical Neurophysiology. 2025;172:S96.

Publications in Vietnamese

1. Ta KV, Nguyen TM, Truong HC, Hoang NTT. Initial assessment of the results and safety of brain angiography procedures at Military Hospital 175. Journal of Military Pharmaco – Medicine. 2014. 6:125-131.
2. Ta KV, Nguyen TM, Truong HC, Hoang NTT. Initial assessment of cerebral vascular intervention procedures at Military Hospital 175. Journal of Practical Medicine. 2014. 937(10):76-79.
3. Ta KV, Bach TT, Phan DV, Hoang NTT. Malignant intracranial dural venous fistula: a case report at Military Hospital 175. Journal of Practical Medicine. 2015. 963(5):65-68.
4. Hoang NTT. Investigation of delayed response by electromyography in patients with sciatic nerve root pain due to disc herniation [Thesis]. Ho Chi Minh, Vietnam: University of Medicine and Pharmacy; 2018.
5. Hoang NTT, Bach TT, Nguyen-Le HT. Investigation of delayed response by electromyography in patients with sciatic nerve root pain due to disc herniation. Ho Chi Minh City Journal of Medicine. 2020. 24(1):161-168.
6. Hoang NTT, Truong CD, Leng M, Nguyen CH. Changes in H reflex in diabetic polyneuropathy. Journal of Military Pharmaco – Medicine. 2020. 2(3):28-32.

7. Truong CD, Ta KV, Phi DN, Hoang NTT, Truong NC. Young cerebral infarction associated with hereditary Protein C deficiency: Report of 1 case. Journal of Military Pharmaco – Medicine. 2020. 2(3):91-97.
8. Truong CD, Hoang NTT. Evaluation of late response in patients with sciatic nerve root pain due to disc herniation by electromyography. Vietnam Medical Journal. 2020. 2:45-50.
9. Truong CD, Hoang NTT. Clinical correlation with abnormal H reflex on electromyography in patients with sciatica due to disc herniation. Clinical Medicine 108. 2020 April 4. 15(4).
10. Vu PT, Tran NT, Lam PAT, Phan HXU, Hoang NTT. Assessing knowledge about stroke of caregivers at Military Hospital 175. Vietnamese Journal of Neurology. 2022. 32:66-71.
11. Duong HTT, Hoang NTT, Nguyen NT, Nguyen HT, Lam PTA, Nguyen PD. Percentage of patients with post-stroke pain and factors in nursing care. Medical – Military Pharmacology. 2022. 1(3):106-112.
12. Pham NTT, Hoang NTT, Tạ KV, Nguyen NH, Phan HXU, Ha DM. The effectiveness of intravenous thrombolytic in treating acute ischemic stroke at Military Hospital 175. Journal of 175 Practical Medicine and Pharmacy. 2022 March. 29(7):72-84.
13. Hoang NTT, Truong NC, Phan HXU. Rate of depression, the relationship between ischemic area and depression in ischemic stroke patients at the military hospital 175. Journal of Military Pharmaco – Medicine. 2022. 5:110-118. doi: 10.56535/jmpm.V20220511
14. Nguyen LN, Huynh LD, Hoang NTT. Posterior reversible encephalopathy syndrome (PRES) and Reversible cerebral vasoconstriction syndrome (RCVS). Vietnamese Journal of Neurology. 2022. 33:52-63.
15. Hoang NTT, Truong NC, Phan HXU. Rate of depression and some related factors in ischemic stroke patients at Military Hospital 175. Vietnamese Journal of Neurology. 2022. 34:30-34..
16. Phan HXU, Huynh LD, Hoang NTT. Management of cerebral edema, cerebral compression and increased intracranial pressure. Vietnamese Journal of Neurology. 2022. 35:53-73.
17. Phan HXU, Hoang NTT. Transcranial Doppler ultrasound: from theory to practice. Vietnamese Journal of Neurology. 2023. 36:49-63.
18. Phan QX, Huynh LD, Hoang NTT Determination of Brain Death/Death by Neurologic Criteria. Vietnamese Journal of Neurology. 2023. 36:68-84.
19. Phan HXU, Doan VAV, Hoang NTT. Clinical epidemiological features of benign paroxysmal postural vertigo: utilizing videonystagmography in Department of Neurology. Vietnamese Journal of Neurology. 2023. 37:10-15.
20. Huynh YNT, Huynh LD, Doan VAV, Hoang NTT. Enhanced detection of discreet focal cortical dysplasia with MRI HARNESS protocol on MRI 3 Tesla: A case report. Vietnamese Journal of Neurology. 2023. 37:20-24.
21. Ly DM, Vo DN, Hoang NTT. Evaluating the effectiveness of spinal pain treatment with repetitive peripheral magnetic stimulation at Military Hospital 175. . Vietnamese Journal of Neurology. 2023. 38:13-16.
22. Phạm T, Hoang N, Tạ V, Nguyễn H, Phan X, Hà M. Đánh giá hiệu quả điều trị đột quỵ nhồi máu não cấp bằng thuốc tiêu sợi huyết đường tĩnh mạch tại bệnh viện quân y 175. Tạp chí Y Dược Thực hành 175. 2023:12.
23. Ly M, Vo N, Hoang N. Đánh giá hiệu quả điều trị đau cột sống bằng kích thích từ trường ngoại biên lặp lại tại Bệnh viện Quân y 175. Tạp chí thần kinh học Việt Nam. 2024;3:13-6.
24. Huynh T, Nguyen X, Nguyen N, Hoang N. Diagnosis and treatment challengesin Multiple Sclerosis with very-Late onset: A case report. Tạp chí thần kinh học Việt Nam. 2024:14-9.

25. Phạm T, Mai T, Le M, Hoang N. Comparison of the value of transcranial ultrasound and brain magnetic resonance in the diagnosis of intracranial artery stenosis due to atherosclerosis in patients with acute cerebral infarction at Military Hospital 175. Tạp chí thần kinh học Việt Nam. 2024:41-51.
26. Huynh T, Nguyen X, Nguyen N, Hoang N. Observation of vitamin D levels in newly diagnosed multiple sclerosis patients: A case series. Tạp chí thần kinh học Việt Nam. 2024:66-9.

Researchs within 3 years

1. Effects of cognitive training application BrainTrain on patients with mild cognitive impairment and apply AI in intervention effectiveness prediction.
2. Building big data of Vietnamese video electroencephalography for smart control utilization and initial application in motor rehabilitation on stroke patients.
3. Genetic landscape of Charcot–Marie–Tooth disease in Vietnam: A prospective multicenter study
4. Comparison of the value of transcranial ultrasound and brain magnetic resonance in the diagnosis of intracranial artery stenosis due to atherosclerosis in patients with acute cerebral infarction at Military Hospital 175
5. Motor-evoked potentials induced by transcranial magnetic stimulation in healthy people at Military Hospital 175
6. First-onset seizure in people over 50 years old: Clinical and electroencephalographic aspects in Military Hospital 175
7. Detection of abnormal EEG findings in patients with epilepsy using natural sleep EEG in the Clinical Neurophysiology Center at Military Hospital 175
8. Prevalence and classification of the horizontal semicircular canal benign paroxysmal positional vertigo in Vietnam. Clinical Neurophysiology
9. Clinicoepidemiology of Benign Paroxysmal Positional Vertigo: Initial Utilizing Videonystagmography in the Department of Neurology

Dr Halle Quang

BSc *Psych&Edu* (First Class Honours), MSc *Brain and Mind Sciences,* PhD *Neuropsychology*
School of Psychology & Brain and Mind Centre
University of Sydney, Australia
E: halle.quang@sydney.edu.au

Executive Summary

I completed my doctoral research training in Neuropsychology at the University of New South Wales, where I investigated apathy in Vietnamese patients with traumatic brain injury. My current research extends to examining dementia-related cognitive and behavioural changes in low and middle-income countries and culturally diverse populations. By employing clinical and neuroimaging methods, I explore the complex interplay between neurobiological and socio-cultural factors underlying disease-specific syndromes, ultimately offering insights for designing culturally appropriate interventions.

Education And Qualifications

2019-2023 **PhD (Neuropsychology)**, University of New South Wales, Australia.

2018 **MSc (Brain and Mind Sciences)**, The University of Sydney, Australia.

2010-2014 **BSc + Hons (Psychology and Education)**, Ho Chi Minh City University of Education, Vietnam (First Class Honours and University Medal)

Academic Appointments

Postdoctoral Research Fellow (full-time) – July 2023 – now
Frontotemporal Dementia Group, School of Psychology and Brain and Mind Centre, University of Sydney, Australia.

Research Officer (full-time) – June 2021 – July 2023
Frontotemporal Dementia Group, School of Psychology and Brain and Mind Centre, University of Sydney, Australia.

Research Assistant (part-time) – Jan 2012 – Nov 2015
School of Psychology, Ho Chi Minh City University of Education, Vietnam.

Awards

- **Best poster –** EMCR/Staff Category, Forefront Scientific Meeting, 2022.
- **PhD Research Award,** International Neuropsychological Society, 2021 (5 recipients worldwide).
- **University Medal,** Ho Chi Minh City University of Education, Vietnam, 2014.
- **Author prize at the National festival for Student Research,** Ministry of Education and Training, Vietnam, 2013.
- **Young Research Talents Award,** Ministry of Education and Training, Vietnam, 2012.
- **Euréka Scientific Research Award for Students,** Ho Chi Minh City Youth Union, Vietnam, 2012.

Research grants and Fellowships

- Travel Grant – **International Brain Research Organisation** to attend to **the International Neuropsychological Society Conference** in San Diego - USA, 2023

- Travel Grant – **International Society of Frontotemporal Dementia Paris-Lille Congress**, 2022

- Sydney Vietnam Institute and Sydney Southeast Asia Centre Seed Funding Grant, 2022 for a research project "*Characterising behavioural and psychological symptoms in Vietnamese older adults: Insights for culturally appropriate research and clinical practice*".

- Australian Graduate Women Hale Barbara Fellowship, 2021 (2 recipients nationwide) for a research project "*Investigating clinical profiles and neurobiological mechanisms of apathy, empathy and depression in dementia*".

- UNSW Scientia PhD and Australian Government Research Training Program Scholarship, 2019 - 2023.

- Scholarship for Outstanding Students, Ho Chi Minh City University of Education, Vietnam, 2010-2014.

- Lawrence S. Tings Scholarship for Excellent Students, Lawrence S. Tings Foundation, Vietnam, 2013 - 2014.

Publications

Journal Articles/Book Chapter	
2025	Li, R., **Quang, H**., Filipčíková, M., Xu, Y., Kumfor, F., Spehar, B., & McDonald, S. (2025). Why Do Cultures Affect Facial Emotion Perception? A Systematic Review. *Journal of Cross-Cultural Psychology*, https://doi.org/10.1177/002202212513348
2025	Mai, L., **Quang, H.,** Nguyen-Martinez, A., & Nguyen, M. (2025). Assessment of Adults and Older Adults. In *How to Support the Neuropsychological Health of the Vietnamese Diaspora* (pp. 66-86). Routledge.
2025	**Quang, H**., Hellerslia, V., Graves, L., & Thieu, Y. (2025). Health-Related Beliefs, Values, and Attitudes. In *How to Support the Neuropsychological Health of the Vietnamese Diaspora* (pp. 21-33). Routledge.
2025	**Quang, H.,** Nguyen-Martinez, A., Nghia, H. T. T., & Mai, L. (2025). The Comprehensive Neuropsychological Interview: Navigating Questions, Interpreters, Collateral Information. In *How to Support the Neuropsychological Health of the Vietnamese Diaspora* (pp. 37-48). Routledge.
2024	Nguyen-Martinez, A. L., Pham, N., Ba, C., Veeramuthu, V., & **Quang, H.** (2024). Pediatric neuropsychological assessment in Southeast Asia: Current status and future directions with Vietnam as a scoping review case example. *Archives of Clinical Neuropsychology*. https://doi.org/10.1093/arclin/acae106
2024	**Quang, H.,** Wearne T., Filipcikova, M., Pham, N., Nguyen, N. & McDonald S. (2024). A biopsychosocial framework for apathy following moderate to severe traumatic brain injury: A systematic review and systematic review. *Neuropsychology Review. 34*(4), *1213-1234.*
2023	Nguyen, M.-N., Pham, R., Nguyen, TV., Lam-Nguyen, NT., McDonald, S. & **Quang, H**. (2023). Neuropsychiatric Symptoms after Moderate to Severe Traumatic Brain Injury in

	Vietnam: Assessment, Prevalence and Impact on Caregivers. *Journal of International Neuropsychological Society, 29*(10), *984-993.*
2023	**Quang, H.,** Le Heron, C., Balleine, B., Nguyen, T-V., Nguyen, T-Q., Nguyen, M.N., Kumfor, F., & McDonald, S. (2023). Reduced Sensitivity to Background Reward Underlying Motivational Disorders: Insight from Apathy after Traumatic Brain Injury. *Neuroscience, 528,* 26-36.
2023	**Quang, H**., Nguyen, A., Do, C, McDonald, S. & Nguyen, C (2023). Examining the Montreal Cognitive Assessment among healthy Vietnamese adults and patients with traumatic brain injury *The Clinical Neuropsychologist. Special Issue on Asian Neuropsychology, 37*(5)*, 1062-1077*
2023	Nguyen, T-Q., Hoang, C. B. D., Hoang Le, M. D., Anh Vo, N. T., **Quang, H.**, Nguyen, C. M., Goodman, C., Savva, G., Pham, VL., Tran, TT., Vo, V.T. & Ha, T.T.H (2023*)*. Examining the Montreal Cognitive Assessment-Basic (MoCA-B) and Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) among elderly Vietnamese. *The Clinical Neuropsychologist. Special Issue on Asian Neuropsychology, 37*(5)*, 1043-1061*
2023	Filipcikova, M., **Quang, H.,** Cassel, A., Darke, L., Wilson, E., Wearne, T., Rosenberg, H. & McDonald, S. (2023). Exploring neuropsychological underpinnings of poor communication after traumatic brain injury: The role pf apathy, disinhibition and social cognition. *International Journal of Language & Communication Disorders*. Accepted on 2022 Dec 21.
2022	**Quang, H.,** Kumfor, F., Balleine, B., Nguyen, T. V., Nguyen, T. Q., Nguyen, M. N., & McDonald, S. (2022). Contributions of intrinsic and extrinsic reward sensitivity to apathy: Evidence from traumatic brain injury. *Neuropsychology, 36*(8), 791–802.
2022	**Quang, H.**, McDonald, S., Huynh-Le, P., Nguyen, T. V., Le, N. A., Lam-Nguyen, N. T., & Kumfor, F. (2022). Apathy in a high prevalence population of moderate to severe traumatic brain injury: An investigation in Vietnam. *Neuropsychology*, *36*(1), 94-102.
2022	**Quang, H.**, Sin, K. & McDonald, S. (2022). Adaptation, validation and preliminary standardisation of the Frontal Systems Behaviour Scale – Apathy Subscale and the Dimensional Apathy Scale in Vietnamese healthy samples. *Journal of the International Neuropsychological Society, 20(3),* 300-310.
2021	**Quang, H.**, Wong, S., Husain, M., Piguet, O., Hodges, J. R., Irish, M., & Kumfor, F. (2021). Beyond language impairment: Profiles of apathy in primary progressive aphasia. *Cortex,* 139, 73-85.
2015	Nguyen, T. & **Quang, H.** (2015). Factors related to retirement crisis in the elderly living in Ho Chi Minh City. *Journal of Science - Special Issue: Social Sciences and Humanities*, *10(76),* 145-150.
2015	**Quang, H.** & Nguyen, M. C. (2015). Investigating sex knowledge of the blind from 12 to 18 years old at Nguyen Dinh Chieu school for the blind, Ho Chi Minh City. *Journal of Science - Special Issue: Educational Studies, 8(74),* 110-119.

2014	Huynh, S., Mai, H. & **Quang, H.** (2014). Alcoholism among young adults in Ho Chi Minh City. *Journal of Science - Special Issue: Social Sciences and Humanities 55(89).* 173-183.

Published Abstracts	
2023	**Quang, H.,** Mencevski, F., Piguet, O., Husain, M., Landin-Romero, R., & Kumfor, F. (2023). Apathy, empathy and depression in Alzheimer's disease and frontotemporal dementia: Examining the constructs and white matter correlates. *Alzheimer's & Dementia*, *19*, e077102.
2023	Landin-Romero, R., **Quang, H.**, Matis, S., D'Souza, A., Calamante, F., & Piguet, O. (2023). Trajectories of white matter degeneration in frontotemporal dementia: new insights using fixel-based analysis. *Alzheimer's & Dementia*, *19*, e075909.
2023	**Quang, H.,** Nguyen, A., Do, C., McDonald, S., & Nguyen, C. (2023). The Vietnamese Montreal Cognitive Assessment: An Evaluation of Construct Validity and Recommended Cut-off for Cognitive Impairment after TBI. *Journal of the International Neuropsychological Society*, *29*(s1), 592-593.
2022	Marian, O. C., Teo, J. D., **Quang, H.**, Lee, J. Y., Kwok, J. B., Halliday, G., ... & Don, A. S. (2022). Frontotemporal dementia caused by progranulin mutations is characterised by disrupted myelin lipid metabolism. *Journal Of Neurochemistry, 162*, 46.
2021	**Quang, H.,** Kumfor, F., Nguyen, T.V., Nguyen, T.Q, Nguyen, M.N, Balleine, B. & McDonald, S. (2021). Altered intrinsic and extrinsic reward sensitivity underpins apathy: Evidence from moderate-to-severe traumatic brain injury. *Journal of the Neurological Sciences, 429*, 119231.

International Conference Oral Presentations

Year	Details
Talks at International conferences	
2024	Nguyen-Martinez, A., Ba, C., Pham, N. & **Quang, H.** (2024). Current State and Future Directions for Pediatric Neuropsychological Evaluations with Vietnamese Children: A PRISMA Review. *52nd INS NYC Annual North American Meeting*, New York, United States
2024	Veeramuthu, V., Ammaiappan, P., Sakamoto, M., **Quang, H**., Sundaram-Patel, S., Choi, E., Yu-Ling, C., Chi-Cheng, Y., Hom, C., Wen-Yu, C., and Fuji, D., on behalf of the ANA International Liaison Taskforce. (2024). Clinical Neuropsychology in Asia: Variability in Training Models, Barriers and the Need for Standardization of Training Guideline. *52nd INS NYC Annual North American Meeting*, New York, United States
2023	McDonald, S., Filipcikova, M. **Quang, H**., et al. (2023). Is there a relationship between emotion dysregulation and social cognition following Traumatic Brain Injury? *International Neuropsychological Society Conference.* Taiwan.
2023	Landin-Romero, R., **Quang, H**., et al. (2023). Trajectories of white and grey matter degeneration in frontotemporal dementia. *International Neuropsychological Society Conference.* Taiwan.
2023	On, N., … **Quang, H.** (2023). Effort Exertion as an Intrinsic Reward: A Cross-Cultural Investigation in Vietnam and Australia. *International Neuropsychological Society Conference.* Taiwan.
2023	**Quang, H.** et al. (2023). Apathy, empathy and depression in Alzheimer's disease and frontotemporal dementia: Examining the constructs and white matter correlates. *International Neuropsychological Society Conference.* Taiwan.

2023	**Quang, H.**, Nguyen, A., Do, C., McDonald, S. & Nguyen, C. (2023). The Vietnamese Montreal Cognitive Assessment: An Evaluation of Construct Validity and Recommended Cut-off for Cognitive Impairment after TBI. *International Neuropsychological Society Conference.* San Diego - USA.
2023	Li, R., Filipčíková, M., Xu, Y., **Quang, H.**, Kumfor, F. & McDonald, S. (2023). Why Do Cultures Affect Facial Emotion Perception? – A Systematic Review. *International Neuropsychological Society Conference.* San Diego - USA.
2022	**Quang, H.** et al. (2022). Apathy, empathy and depression in behavioural frontotemporal dementia and Alzheimer's disease: Are we talking about the same thing? *International Society of Frontotemporal Dementia Biennial Congress.* Paris – Lille, France.
2022	Marian, O. C., Teo, J. D., **Quang, H.**, Lee, J. Y., Kwok, J. B., Halliday, G., ... & Don, A. S. (2022). Frontotemporal dementia caused by progranulin mutations is characterised by disrupted myelin lipid metabolism. *International Society for Neurochemistry Conference,* Hawaii, USA.
2022	**Quang, H.** et al. (2022). Apathy in a high prevalence population of moderate to severe traumatic brain injury: An investigation in Vietnam. *Southeast Asia Mental Heath Conference.* Online
2022	**Quang, H.** et al. (2022). Contributions of Foreground and Background Reward Sensitivity to Apathy after Traumatic Brain Injury. *International Neuropsychological Society Conference.* Barcelona - Spain.
2022	**Quang, H.** et al. (2022). Examining Apathy in Vietnam – a Country with High Prevalence of Moderate to Severe Traumatic Brain Injury. *International Neuropsychological Society Conference.* New Orleans – USA (virtual).
2021	**Quang, H.** et al. (2021). Altered intrinsic and extrinsic reward sensitivity underpins apathy: Evidence from moderate-to-severe traumatic brain injury. *World Congress of Neurology*. Rome – Italy (virtual).
2021	**Quang, H.** et al. (2021). Do more for more: Reward processing underlying apathy following moderate-to-severe traumatic brain injury. *The 6th Pacific Rim Conference*. Melbourne – Australia (virtual).
Talks at National conferences	
2022	**Quang, H.** et al. (2022). Motivational disorders after traumatic brain injury: Is apathy driven by abnormal sensitivity to foreground and background reward? *Australasian Society for the Study of Brain Impairment Conference.* Perth – Australia (virtual).
2021	Lee, C., **Quang, H.** et al. (2021). White matter changes underlying frontotemporal lobar degeneration pathology. *Brain and Mind Symposium.* Sydney – Australia.
2021	**Quang, H.** et al. (2021). Beyond language impairment: Profiles of apathy in primary progressive aphasia. *Forefront Annual Meeting 2021.* Sydney – Australia.
2021	**Quang, H.** et al. (2021). Characteristics of apathy and depression between Australians and Vietnamese: A cross-cultural study. *Australasian Society for Social and Affective Neuroscience Conference*. Melbourne – Australia (virtual).
2020	**Quang, H.** et al. (2020). Cultural considerations in assessing motivational deficits - insights from the Vietnamese validation of apathy questionnaires. *Sydney Vietnam Symposium*. Sydney – Australia.
2019	**Quang, H.** et al. (2019). Clinical profiles and neural correlates of apathy in primary progressive aphasia. *Frontier Lab Meeting.* Sydney – Australia.
Posters	
2024	Piguet, O., Chuang, T., Matis, S. **Quang, H**. et al. (2024). Grey and White Matter Correlates of Motor Speech Disturbances in Nonfluent Progressive Aphasia Syndromes. *52nd INS NYC Annual North American Meeting*, New York, United States

2023	My-Ngan, N., … **Quang, H.** (2023). No Existence or Under-recognition? Empirical Evidence for the Prevalence and Impact of Neuropsychiatric Symptoms after TBI in Vietnam. *International Neuropsychological Society Conference.* Taiwan.
2022	Landin-Romero, R., Lee, C., **Quang, H.** et al. (2022). Trajectories of brain changes predicting distinct pathology subtypes in frontotemporal dementia. *International Society of Frontotemporal Dementia Biennial Congress.* Paris – Lille, France.
2022	**Quang, H.** et al. (2022). Apathy, empathy and depression in behavioural frontotemporal dementia and Alzheimer's disease: Are we talking about the same thing? *Forefront Annual Meeting 2022.* Sydney – Australia.
2022	Landin-Romero, R., **Quang, H.** et al. (2022). Trajectories of white matter degeneration in frontotemporal dementia: new insights using fixel-based analysis. *Forefront Annual Meeting 2022.* Sydney – Australia.
2021	**Quang, H.** et al. (2021). Unfolding cultural variances of apathy and depression: A comparison between the Vietnamese and Australians. *The 6th Pacific Rim Conference*. Melbourne – Australia.
2021	**Quang, H.** et al. (2021). Influence of Family Functioning, Overprotectiveness and Self-Efficacy in Apathy after Traumatic Brain Injury. *International Neuropsychological Society Conference.* San Diego – USA.
2019	**Quang, H.** et al. (2019). Uncovering emotional apathy in primary progressive aphasia: A clinical and neuroimaging study. *Australasian Society for Social and Affective Neuroscience Conference.* Newcastle – Australia.

Supervision and teaching experience

Supervisor – University of New South Wales and University of Sydney – Jan 2020 - now

- Supervise and mentor Honours (in neuropsychology and speech pathology) and Master of Medicine research projects. All my students successfully completed their theses on time and with First Class Honours. Most of their research findings have been published or are currently in preparation for submission to peer-reviewed journals.

Manager –University of Sydney – Jan 2020 - now

- Oversee research activity of the Neuroimaging lab at FRONTIER – the frontotemporal dementia research clinic at University of Sydney. I provide support and training to a group of 8 students, research assistants and supporting staff.

Visiting lecturer – Hoa Sen University, Vietnam – April 2022 – now

- Taught courses in *biological psychology*, *neuroscience* and *neuropsychology*.
- Developed and implemented course materials, including syllabi, assignments, and assessments.
- Worked with colleagues to develop interdisciplinary courses and initiatives.
- Advised and mentored students on academic and career-related matters.

Visiting lecturer – Ho Chi Minh City University of Technology

- Taught courses in *developmental psychology*, with a focus on neurobiological basis of human behaviour and development.
- Developed and implemented course materials, including syllabi, assignments, and assessments.

Service and Community Engagement

Service

- Scientific Committee, **International Neuropsychology Society Conference**, Taiwan, 2023
- Member of Research Committee, **Asian Neuropsychology Association**, International Neuropsychological Society, 2022 - now.
- Student Representative, **Vietnam Neuropsychology Network**, 2022 - now.
- Student Representative, **Australasian Society for the Study of Brain Impairment,** 2019.
- President, **Student Psychological Society, Ho Chi Minh City University of Education**, 2013.

Community Engagement

- Founder - **A Brainy Chat** - a non-profit organisation that aims to disseminate psychology and neuroscience knowledge to the Vietnamese public in an engaging and interactive way, 2022 – now.
- Invited speaker - **Open talk - Neuroscience for children**, Vietnam, 2020.
- Organiser - **Trivial Night** to promote activities of Australasian Society for the Study of Brain Impairment, University of New South Wales, 2019.
- Volunteer - **Australasian Society for the Study of Brain Impairment Conference,** Wellington, New Zealand, 2019.
- Member - Australasian Society for the Study of Brain Impairment, Australasian Society for Social and Affective Neuroscience and International Neuropsychological Society.

Duc Nguyen Minh Vu

vnmduc.work@gmail.com | +84 975-944-987 | Vu Nguyen Minh Duc | HeigatVu

Professional Summary

A technically skilled researcher with a unique background combining clinical physiotherapy experience and a Master of Biomedical Engineering candidate. Aims to bridge the gap between healthcare and technology by applying machine learning, natural language processing, and speech analysis to create impactful diagnostic tools for neurodegenerative diseases.

Education

M.Sc **International University - Vietnam National University Ho Chi Minh City** — Mar 2025 – Current
Biomedical Engineering

B.Sc **University of Medicine and Pharmacy at Ho Chi Minh City** — Sept 2018 – Aug 2022
Rehabilitation Technique

Experience

City Children's Hospital, Physical Therapy — Ho Chi Minh City, Vietnam, Mar 2023 – Feb 2025

- Collaborated effectively within a multidisciplinary team of doctors, nurses, and other healthcare professionals to manage patient care.
- Communicated complex medical information and patient conditions clearly to both clinical staff and patients' families.

Projects

Sentiment analysis — github.com/HeigatVu/sentiment-analysis

- Developed a complete text classification pipeline, including data preprocessing (NLTK), TF-IDF vectorization, and model training (Decision Tree, Random Forest).
- Tools Used: Python, Pandas, NLTK, Scikit-learn, Matplotlib, Seaborn

LLM reading PDF — github.com/HeigatVu/LLM-reading-PDF

- To build a conversational AI that allows users to query and receive contextually accurate answers from technical PDF documents.
- Tools Used: Python, LangChain, ChromaDB, Huggingface Transformer, Chainlit

Course

Deep Learning Specialization — Dec 2024
DeepLearning.AI

- Completed 5 courses covering CNNs, RNNs, LSTMs, Transformers, and optimization strategies.

coursera.org/account/accomplishments/specialization/AR469QJLXFVF

Machine Learning Specialization — Jul 2024
Stanford University & DeepLearning.AI

- Completed 3 courses covering supervised learning, unsupervised learning, and reinforcement learning.

coursera.org/account/accomplishments/specialization/HM8JZRJDST7G

Mathematics for Machine Learning and Data Science Specialization May 2024

DeepLearning.AI

- Completed 3 courses covering foundational linear algebra, calculus, probability, and statistics for AI.

coursera.org/account/accomplishments/specialization/9MJ8RY5WRNUS

CURRICULUM VITAE

NAME: THU NGOC MINH PHAN
POSITION TITLE: MSc Student
EDUCATION

Institution and Location	Degree	Completion Date	Field of Study
International University, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	M.S.	2022-2025	Biomedical Engineering
University of Science, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	B.S.	2017-2021	Medical Biotechnology

A. Personal Statement and Research Interest

I am a Master's student of Biomedical Engineering at International University, Vietnam National University HCMC. Being a dedicated researcher is the greatest dream of my life. I aspire to contribute significantly to advancing diagnostic and therapeutic methodologies to enhance human health. Thus, I always maintain a positive attitude, fostering my readiness to absorb new knowledge and skills. Working at Brain Health Lab, my research focused on EEG signal processing and data analysis to gain insights into brain activity, specifically targeting interventions for individuals experiencing mild cognitive impairment to prevent the progression to dementia in the early stages. In the future, I look forward to learning about exploiting the potential of artificial intelligence (AI) in neuro-science and neuro-engineering for clinical use.

B. Current and Past Positions

(02/2022 – present) Research assistant at Brain Health Lab, School of Biomedical Engineering, International University, Ho Chi Minh City - Vietnam National University HCM City, Vietnam.

- Main research direction: Focusing on early intervention methods for Alzheimer's disease, with Master's thesis titled "Effects of cognitive training games in Brain Train application on Mild Cognitive Impairment patients and applying Machine Learning models to predict intervention effect."
- Participating in other lab projects, including Brain-computer interface (02/2022 - 02/2024), Depression and anxiety studies (06/2023 - 02/2024).
- (2023) Deputy organizing committee of the BrainConnects 2023 conference with the theme "Healthy Brain Aging: from Technology to Intervention."
- Supporting my supervisor in guiding students in scientific research projects for academic courses (Project 1, Project 2, Pre-thesis).
- Supporting my supervisor (as teaching assistant) for Master's class in "Research Methodology" (Semester 1, 2023-2024), for PhD class in "Biostatistics" (Semester 2, 2024-2025).
- Being provided opportunities to fully develop scientific capabilities, including writing research proposals, leading projects, conducting experiments, performing analysis and interpretation, publishing research findings, preparing posters and presentations for participation in national and international conferences, attending training programs, seminars, webinars, and workshops.

(06/2020 – 10/2021) Trainee at Stem Cell Institute, University of Science, Ho Chi Minh City - Vietnam National University HCM City, Vietnam.

- Conduct Bachelor's thesis with the topic "Evaluation of the immunomodulatory properties and cartilage differentiation capability of umbilical cord mesenchymal stem cells in inflammatory microenvironment in vitro."

- Cultivating and expanding mesenchymal stem cells (MSCs), and establishing co-culture systems for inflammatory studies.
- Differentiating MSCs into bone, cartilage, and fat cells, and analyzing cell morphology.
- Measuring cytokine levels and performing qPCR analysis to evaluate cartilage expression genes.

C. Awards and Honors

(2025) Second Prize of the 5th Vietnam Medical Achievement Award, Vietnam.

(2024) First Prize, Scientific Conference "Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment", Vietnam.

(2024) IBRO-APRC Associate School on "Neuro-Enhancement for Brain Health", Department of Physiology, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand.

(2023) Research Assistant Fellowship, School of Biomedical Engineering, International University, Ho Chi Minh City - Vietnam National University, Ho Chi Minh City, Vietnam.

(2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.

(2023) Third Prize in the contest "Digital Transformation - Creativity Challenge", Ho Chi Minh City, Vietnam.

(2023) ADAI Fellowship, Ho Chi Minh City, Vietnam.

(2022) ITR Graduate Fellowship, Ho Chi Minh City, Vietnam.

D. Publications, Presentations and Posters

<u>Publications:</u>

Assessing the feasibility of cognitive intervention via BrainTrain cognitive gaming application for Mild Cognitive Impairment patients (Accepted, 2025, International

Journal of Bioinformatics Research and Applications). Authors: **Thu Ngoc Minh Phan**, Binh Nguyen Le Nguyen, Bao Quoc Nguyen, Nhung Hoang Do, Nhu Thi Huynh Tran, Nhan Thanh Nguyen, Thu Hoang Anh Pham, Nhan Quoc Trung Nguyen, Tuong Huu Nguyen, Kien Trung Duong, Nghia Tien Trong Hoang, Thu Thi Hoai Tran, Dang Minh Ly, and Dieu Xuan Nguyen Tan Le-Duy, and Huong Ha Thi Thanh.

Nguyen, M. T. D., Xuan, N. Y. P., Pham, B. M., Do, H. T. M., **Phan, T. N. M.**, Nguyen, Q. T. T., ... & Ha, H. T. T. (2023). Optimize temporal configuration for motor imagery-based multiclass performance and its relationship with subject-specific frequency. *Informatics in Medicine Unlocked*, *36*, 101141.

Truong, N. C., **Phan, T. N. M.**, Huynh, N. T., Pham, K. D., & Van Pham, P. (2023, June). Interferon-Gamma Increases the Immune Modulation of Umbilical Cord-Derived Mesenchymal Stem Cells but Decreases Their Chondrogenic Potential. In *International Conference Innovations in Cancer Research and Regenerative Medicine* (pp. 19-33). Cham: Springer International Publishing.

Phan, T. N. M., Nguyen, H. T., Huynh, T. N. M., Nguyen, T. H., Tran, T. N. Y., & Ha, H. T. T. (2022, December). Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 719-737). Cham: Springer Nature Switzerland.

<u>Presentations:</u>

(2024) 14th Mind-Body Interface Internationals Symposium, China Medical University, Taichung, Taiwan: “Assessing the Feasibility of Cognitive Intervention via BrainTrain Cognitive Gaming Application for Mild Cognitive Impairment Patients.”

(2024) Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment”: “Development and testing of the BrainTrain cognitive training application for Mild Cognitive Impairment (MCI) patients.”

(2023) Hyper-Interdisciplinary Conference 2023 (Flash talk): “Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly.”

(2022) The 9^{th} International Conference in Vietnam on the development of Biomedical Engineering: "Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly."

<u>Posters:</u>

(2024) International Conference on Neuroscience and Brain Health 2024 (ICNB2024) and the 27th Thai Neuroscience Society Annual Conference (TNS27): "Assessing the Feasibility of Cognitive Intervention via BrainTrain Cognitive Gaming Application for Mild Cognitive Impairment Patients."

(2024) Scientific Conference "Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment": "Development and testing of the BrainTrain cognitive training application for Mild Cognitive Impairment (MCI) patients."

(2021) BrainConnect Conference 2021: "Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly."

E. References

Huong Thi-Thanh Ha, Ph.D.

Head of Brain Health Lab

Chair, Department of Tissue Engineering and Regenerative Medicine

School of Biomedical Engineering, International University

htthuong@hcmiu.edu.vn

Nhat Chau Truong, M.S.

Head of Center for Biomedical Training

Stem Cell Institute, University of Science

nhattruong@sci.edu.vn

Duc Q. Nguyen

Email: nguyenquangduc2000@gmail.com — nqduc@hcmut.edu.vn
Website: https://martinakaduc.github.io

D.O.B: December 24^{th}, 2000
Phone: (+84) 898 986 370

RESEARCH SUMMARY

Research Interests:
- Generative Models
- Graph Representation Learning
- Probabilistic Machine Learning

Application Domains:
- Healthcare
- Education
- Natural Language Processing

EDUCATION

Bachelor of Engineering in Computer Science *Aug. 2018 - Jun. 2022*
- University of Technology - Vietnam National University HCMC, Vietnam
- Cumulative GPA: 3.50/4.00, First Class Honor
- Thesis: AI-Powered Decision Support System for Antiviral Pharmaceutical Formulation
- Thesis Grade: 3.84/4.00 (9.6/10)

Master of Engineering in Computer Science *Jan. 2023 - Jan. 2025*
- University of Technology - Vietnam National University HCMC, Vietnam
- Cumulative GPA: 3.65/4.00
- Thesis: Optimizing subgraph isomorphism prediction models with application orientation to the drug design process
- Thesis Grade: 3.72/4.00 (9.3/10)

WORKING EXPERIENCE

Teaching Assistant *Apr. 2023 - now*
- University of Technology - Vietnam National University HCMC

RESEARCH EXPERIENCE

Visiting Student Researcher - Stanford University
- Stanford Trustworthy AI Research Lab *Sep. 2024 - Dec. 2024*

Research Assistant
- Vietnamese-Swiss Joint Research Project *Feb. 2022 - Dec. 2023*
Project: "Human and Algorithms for Detecting and Counter-attacking Fake Medical News"
- Vietnam National Scientific Project *Feb. 2022 - Sep. 2025*
Project: "Building an automatic translation system from Vietnamese to Bahnar and a Bahnar speech synthesis system (including all dialects)" – Role: Member

PUBLICATIONS

Conference papers:

1. Sang T. Truong, Duc Q. Nguyen, Willie Neiswanger, Ryan-Rhys Griffiths, Stefano Ermon, Nick Haber, Sanmi Koyejo, "Nonmyopic Bayesian Optimization in Dynamic Cost Settings," in *ICLR Workshop on Agentic AI for Science: Hypothesis Generation, Comprehension, Quantification, and Validation*, (**Workshop at A*-ranked Conference**), 2025.
2. Sang T. Truong*, Duc Q. Nguyen*, Toan Nguyen*, Dong D. Le*, Nhi N. Truong*, Tho Quan, Sanmi Koyejo, "Crossing Linguistic Horizons: Finetuning and Comprehensive Evaluation of Vietnamese Large Language Models," in *Proceedings of The 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics* (**NAACL, A-ranked Conference**), 2024.
3. Khang N. H. Vo, Dong Le, Dat M. T. Phan, Sang T. Nguyen, Nguyen Q. Pham, Oanh N. Tran, Duc Q. Nguyen, Hieu M. T. Vo, Tho Quan, "Revitalizing Bahnar Language through Neural Machine Translation: Challenges, Strategies, and Promising Outcomes," in *The 14th Symposium on Educational Advances in Artificial Intelligence* (**EAAI, Symposium at A*-ranked Conference**), 2024.
4. Nhi Ngoc Truong, Duc Quang Nguyen, Jeffrey Gropp, Sang T. Truong, "Hybrid Transformer and Holt-Winter's Method for Time Series Forecasting," *ICLR Workshop: Learning from Time Series for Health* (**Workshop at A*-ranked Conference**), 2024.
5. Sang T. Truong, Duc Q. Nguyen, Tho Quan, Sanmi Koyejo, "Thomas: Learning to Explore Human Preference via Probabilistic Reward Model," in *ICML Workshop: The Many Facets of Preference-Based Learning* (**MFPL, Workshop at A*-ranked Conference**), 2023.
6. Thanh Toan Nguyen, Quang Duc Nguyen, Zhao Ren, Jun Jo, Quoc Viet Hung Nguyen, Thanh Tam Nguyen, "10X Faster Subgraph Matching: Dual Matching Networks with Interleaved Diffusion Attention," In *Proceedings of 2023 International Joint Conference on Neural Networks* (**IJCNN, B-ranked Conference**), 2023.
7. Hoang-Danh Nguyen, Thanh-Phong To, Duc Q. Nguyen, Thanh-Trung Huynh, Tham Tran, Cong-Tuan Bui, Tho Quan, "A novel approach for non-query fake news detection using K-SOM and Graph

Neural Networks," in *Proceedings of* 15^{th} *International Conference on Knowledge and Systems Engineering* (KSE), 2023.

8. Duc Q. Nguyen, Khoan D. Le, Bach T. Ly, An D. Nguyen, Quang H. Nguyen, Tuan H. Nguyen, Cuong Quoc Duong, Thanh N. Truong, Phuong Thuy Viet Nguyen, Tho T. Quan, "Towards *de Novo* Drug Design for the Coronavirus: A Drug-Target Interaction Prediction Approach Using Atom-Enhanced Graph Neural Network with Multi-Hop Gating Mechanism," In *Proceedings of* 9^{th} *NAFOSTED Conference on Information and Computer Science* (**NICS**), 2022. **Best Paper Award.**
9. Cuong Nguyen Dang, Minh Nguyen Huynh, Duc Nguyen Quang, Duc Nguyen Quang, Thinh Nguyen Tien, Khuong Nguyen An, Chon Le Trung, and Tho Quan Thanh, "BeCaked+: An Explainable AI Model to Forecast Delta-spreading Covid-19," In *Proceedings of 2022 International Conference on Innovations in Computing Research* (ICR'22), 2022.

Journal articles:

1. Duc Q. Nguyen, Thanh Toan Nguyen, Jun Jo, Florent Poux, Shikha Anirban, Tho Quan, "Explainable Neural Subgraph Matching with Learnable Multi-hop Attention," *IEEE Access* (**Q1**), 2024
2. Duc Q. Nguyen, Nghia Q. Vo, Thinh T. Nguyen, Khuong A. Nguyen, Quang H.Nguyen, Dang N. Tran, and Tho T. Quan, "BeCaked: An Explainable Artificial Intelligence Model For COVID-19 Forecasting," *Scientific Reports* (**Q1**), vol. 12, no. 7969, 2022.
3. Duc Nguyen, Thien Pham, and Tho Quan, "Design, implementation and evaluation for a high precision prosthetic hand using MyoBand and Random Forest algorithm," *Science & Technology Development Journal - Engineering and Technology*, vol. 3, pp. 28-39, 2020.

FELLOWSHIPS, SCHOLARSHIPS

Master student:

- *Microsoft Accelerate Foundation Models Research (AMFR)*
$60.000 USD — *Nov. 2023 - Jan. 2025*
- *Master, PhD Scholarship Programme of Vingroup Innovation Foundation (VINIF)*
$5.000 USD — *Dec. 2022 - Nov. 2023*

Undergraduate:

- *Honda Award 2020* - $450 USD
- *VinAI Scholarship 2020* - $450 USD
- *KSYS-CUBE Scholarship 2020* - ¥50.000 JPY
- *Odon Vallet Fellowship 2018* - $425 USD
- *Odon Vallet Fellowship 2017* - $400 USD

MANAGEMENT AND ORGANIZING EXPERIENCE

Organizer — *Dec. 2024 - now*
- *NAACL 2025 Workshop on Language Models for Underserved Communities*

Laboratory Senior Member — *Jun. 2022 - now*
- *VNPT-HCMUT Laboratory*

Project Manager — *Nov. 2019 - Nov. 2021*
- *Bach Khoa Artificial Intelligence Club (BKAIC), University of Technology - VNU-HCM*

CERTIFICATES, MERITS, AND PRIZES

First Prize in Biomedical - Euréka Student Scientific Research Award — *Apr. 2021*
- *Minister of the Ministry of Information and Communications*

Certificate of Merit: Students with excellent achievements in learning and research in the field of Information Technology, Artificial Intelligence — *Dec. 2019 and Dec. 2021*
- *Director of the HCMC Department of Information and Communication*

Bach Khoa Youth Award 2020: Certificate of Typical Student — *Dec. 2020*
- *University of Technology - Vietnam National University HCMC*

REFERENCES

Associate Professor Tho Quan
- *Dean of Faculty of Computer Science and Engineering, University of Technology - VNU-HCM*
- *Email:* qttho@hcmut.edu.vn

Assistant Professor Sanmi Koyejo
- *Department of Computer Science - Stanford University*
- *Email:* sanmi@stanford.edu

Dr. Thanh-Toan Nguyen
- *School of Information and Communication Technology - Griffith University*
- *Email:* thanhtoan.nguyen@griffith.edu.au